

NORTHCAPE REFINING COMPANY (FICTIONAL)

Ravensfell Refinery · Unit 22, Crude Distillation · Mechanical Integrity Department

STATIC EQUIPMENT EXTERNAL INSPECTION REPORT

Report no.	NRC/RVF/INSP/2026-0417	Procedure	NRC-INS-STD-014 Rev 3
Inspection date	12 August 2026	Unit	22 — Crude Distillation
Inspector	R. Kulkarni, Level II (fictional)	Reviewed by	S. Marathe, Inspection Engineer (fictional)
Inspection type	External, in service	Next due	August 2028

1. Scope

This report covers the routine external inspection of static equipment and associated piping in Unit 22 carried out on 12 August 2026 with the unit in service. Visual examination was supplemented by ultrasonic thickness readings at the condition monitoring locations listed in section 4. No internal examination was performed and no equipment was opened. Findings are recorded against the tag numbers held in the plant register.

2. Equipment covered

Tag	Description	Design press. (barg)	Design temp. (°C)	Last external
V-2201	Overhead accumulator, crude column	9.5	120	2024-07
E-1104A	Crude / residue exchanger, channel end	24.0	310	2023-11
P-3302B	Reflux pump, standby	16.0	95	2025-02
PSV-2207A	Relief valve, accumulator outlet	9.5	120	2024-04
TK-0450	Slop tank, fixed roof	Atmospheric	60	2022-09

3. Findings

No.	Tag	Observation	Severity	Recommended action
1	V-2201	Paint blistering on the south shell over roughly 0.4 m², adjacent to nozzle N3. No metal loss visible.	Minor	Clean and recoat at the next shutdown.
2	E-1104A	Insulation cladding open at the channel end. Corrosion under insulation suspected over a 1.2 m run.	Major	Strip cladding and UT scan before restart.
3	PSV-2207A	Test tag expired 04-2026. No current bench certificate held against this tag in the register.	Major	Withdraw and bench test. Fit spare.
4	P-3302B	Mechanical seal weeping at the outboard face, approximately three drops per minute.	Minor	Monitor weekly and plan a seal change.
5	L-2210	Support shoe displaced 18 mm at pipe rack bent 14. Line resting on the guide edge.	Minor	Reset the shoe and verify hanger load.

scaffold needed before UT

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4. Thickness readings

Readings taken with a calibrated digital thickness gauge, couplant applied, at the condition monitoring locations held in the plant register. Values in millimetres.

CML	Location	Nominal	Measured	Min. allowable	Status
01	V-2201 shell, course 1	12.00	11.42	9.60	Accept
02	V-2201 shell, course 2	12.00	11.08	9.60	Accept
03	V-2201 bottom head	14.00	12.91	10.20	Accept
04	E-1104A channel, 3 o'clock	16.00	13.05	12.70	Watch
05	E-1104A shell, mid course	16.00	15.44	12.70	Accept
06	L-2210 elbow, 6 o'clock	9.27	7.16	6.90	Watch

5. Conclusion

Unit 22 static equipment remains fit for continued service until the next scheduled shutdown, subject to the two major findings in section 3 being closed out. CML 04 on E-1104A and CML 06 on L-2210 are within 0.5 mm of their minimum allowable thickness and are placed on a six-monthly reading interval. A non-conformance report is to be raised against PSV-2207A, which is operating outside its certification interval.

*Agreed - raise NCR-114 and
re-inspect after Nov S/D.*

Inspector — R. Kulkarni (fictional)

Reviewed — S. Marathe (fictional)

14/08/26